

4.2.5 Anti-Aliasing (AA)

Aliasing refers to the jagged edges seen on slanting lines in any rendered image. Anti-Aliasing is the process of removing or reducing the effects of Aliasing. AA achieves smoother lines by taking a sample of the pixels around the point and using this information to fill the jagged edges. The process can take different forms, like FSAA or Full Screen Anti Aliasing, where all the pixels on the screen are sampled before making remedial modifications to the image. FSAA can also offer better result if more samples were taken. Depending on the number of samples taken, AA is denoted as 2X (2 samples), 4X, 8X, and 16X. Needless to say, taking these samples requires processing power, and usually the higher sample rates are achieved only by powerful graphics cards.



An unantialiased frame



A 4x antialiased frame

4.2.6 Artefacts

Artefacts are the defects that appear in a display. They are usually the result of bottlenecks in the system rather than defects in the display unit. Jagged edges and pixelated images are examples of artefacts.

4.2.7 Bilinear Filtering

To smoothen the texture at a particular point, a sample of values of the texels around the point is taken. This process is called Bilinear Filtering, and is used to arrive at values within one mipmap.

4.2.8 Bump Mapping

A bump map is a set of values that contain information about the elevation of points on an object's surface. After a texture has been mapped, to increase realism, its surface can be modified according to the values in the bump map. This process is called bump mapping.

4.2.9 Core clock

This refers to the operating frequency of the GPU of the graphics card. Like a CPU, the GPU, too, can be overclocked.